

Expressivity

Which functions can a given neural network express / approximate?

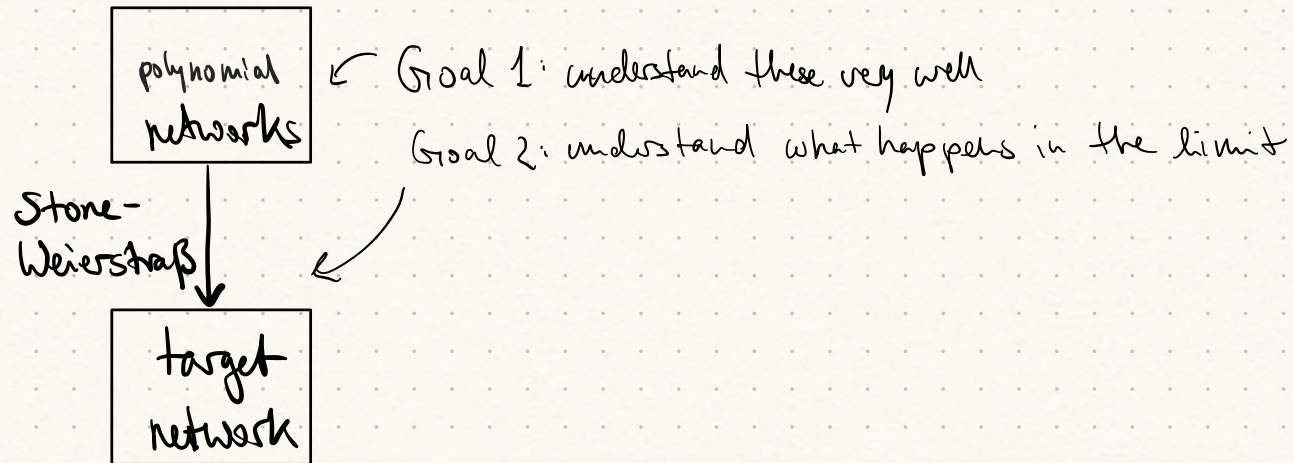
Training

Why can we find "good" critical points?

Generalization

Why does the found solution function work well on new data?

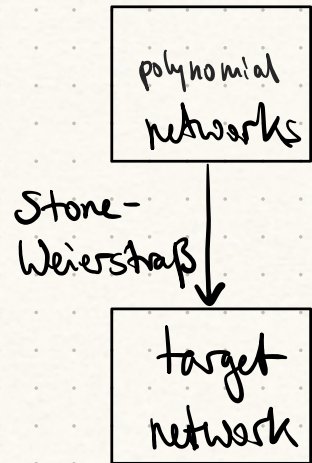
AG approach to deep learning theory



Algebra Unveils Deep Learning An Invitation to Neuroalgebraic Geometry

Giovanni Luca Marchetti^{*1} Vahid Shahverdi^{*1} Stefano Mereta^{*1} Matthew Trager^{*2} Kathlén Kohn^{*1}

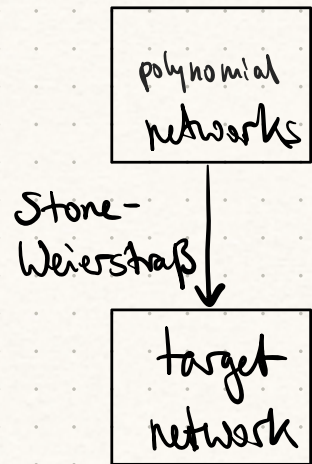
AG approach to deep learning theory



Upside:

- ① Can approximate arbitrary networks
- ② tractable via AG tools & next natural step beyond linear nets

AG approach to deep learning theory



Upshots:

- ① Can approximate arbitrary networks
- ② tractable via AG tools & next natural step beyond linear nets
- ③ neuromanifolds in *finite-dimensional* ambient spaces

For a MLP,

its neuromanifold lives in a finite-dimensional vector space

if and only if

it is polynomial.

(Reason: Universal Approximation Theorem)

Universal Approximation Theorem

Leshno, Lin, Pinkus, Schocken: Multilayer feedforward networks with a non-polynomial activation function can approximate any function.
Neural Networks 6, 1993:

Theorem 1:

Let $\sigma \in M$. Set

$$\Sigma_n = \text{span} \{ \sigma(w \cdot x + \theta) : w \in R^n, \theta \in R \}.$$

Then Σ_n is dense in $C(R^n)$ if and only if σ is not an algebraic polynomial (a.e.).

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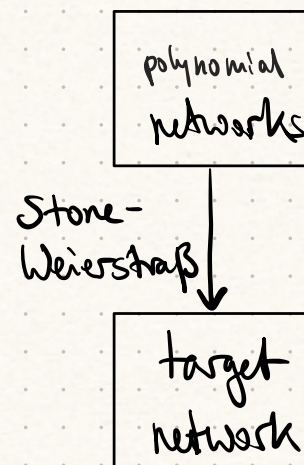
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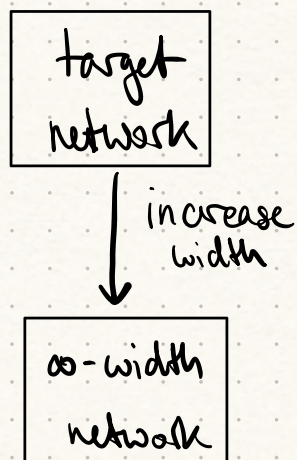
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polynomials are the choice
to approximate networks with
models in finite-dimensional spaces

AG approach

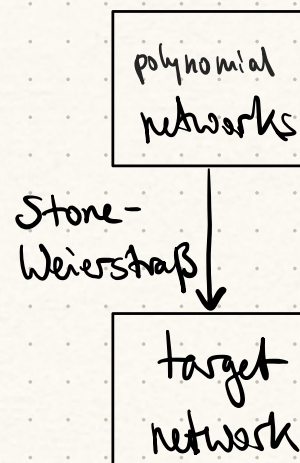


neural
tangent
kernel
NTK approach



linearized models
of ∞ dimension

algebraic
geometry
AG approach



semi-algebraic models in
finite-dimensional ambient spaces

Neural Tangent Kernel: Convergence and Generalization in Neural Networks

Arthur Jacot, Franck Gabriel, Clement Hongler

Advances in Neural Information Processing Systems 31 (NeurIPS 2018)

> 4000 citations

Expressivity

Which functions can a given neural network express / approximate?

Training

Why can we find "good" critical points?

Generalization

Why does the found solution function work well on new data?

Understand neuromanifold:

dimension,

geometry (boundary, singularities, ...),

ϵ -tube,

...

Network training = 'distance' minimization

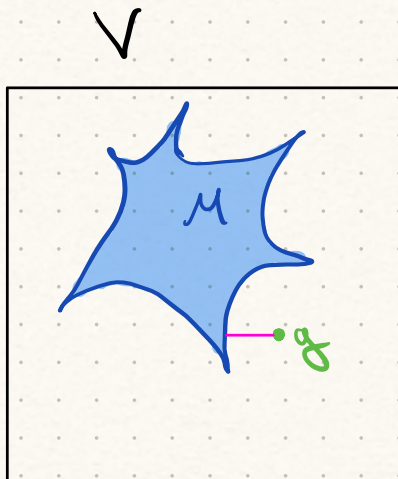
Let $\mathcal{M} \subseteq V := \left(\mathbb{R}[x_1, \dots, x_{d_0}] \leq \mathbb{D} \right)^{d_L}$,
 $\nwarrow$ neuromanifold

$S \subseteq \mathbb{R}^{d_0} \times \mathbb{R}^{d_L}$ finite dataset,

$\nwarrow$ mean squared error
 MSE loss: $\mathcal{L}(f) := \sum_{(a,b) \in S} \|f(a) - b\|^2$

$\nwarrow$ [dist(f,g) = 0 possible for $f \neq g$]

Proposition: There is a pseudometric $\text{dist}: V \times V \rightarrow \mathbb{R}_{\geq 0}$ and some $g \in V$ such that minimizing $\mathcal{L}(f)$ over $f \in \mathcal{M}$ is equivalent to minimizing $\text{dist}(f, g)$ over $f \in \mathcal{M}$.



**Critical Points of Degenerate Metrics
on Algebraic Varieties:
A Tale of Overparametrization**

Prop.
2.6

Giovanni Luca Marchetti *
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Erin Connelly *
University of Osnabrück

Paul Breiding *
University of Osnabrück

Kathlén Kohn *
KTH Royal Institute of Technology
& Digital Futures

Network training = 'distance' minimization

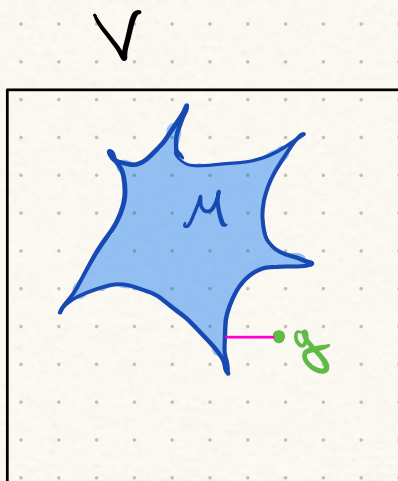
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$\nwarrow$ ∇ not generic

"multivariate Hankel type"

Loss Landscape

$$= \{(\theta, \mathcal{L}(f_\theta)) \mid \theta \in \Theta\}$$

params input output
↓ ↓ ↓
 $\mu: \Theta \times X \rightarrow Y$
 $\mathcal{M} = \{ \underbrace{\mu(\theta, \cdot)}_{=f_\theta} \mid \theta \in \Theta \}$

Loss Landscape

$$= \{(\theta, \mathcal{L}(f_\theta)) \mid \theta \in \Theta\}$$

$$\begin{array}{ccc} \text{params} & \text{input} & \text{output} \\ \downarrow & \downarrow & \downarrow \\ \mu: \Theta \times X & \rightarrow & Y \\ \mathcal{M} = \{ \underbrace{\mu(\theta, \cdot)}_{=f_\theta} \mid \theta \in \Theta \} \end{array}$$

can be studied in a decoupled way:

$$\begin{array}{ccccc} \Theta & \xrightarrow{\quad} & \mathcal{M} & \xrightarrow{\mathcal{L}} & \mathbb{R} \\ \theta & \mapsto & f_\theta & & \end{array}$$



loss landscape in function space:

$$= \{(f, \mathcal{L}(f)) \mid f \in \mathcal{M}\} \subseteq V \times \mathbb{R}$$

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Geometry of $\mathcal{M}$ affects loss landscape!

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Why can we find "good" critical points?

Understand loss landscape:

count & classify crit. pts.,

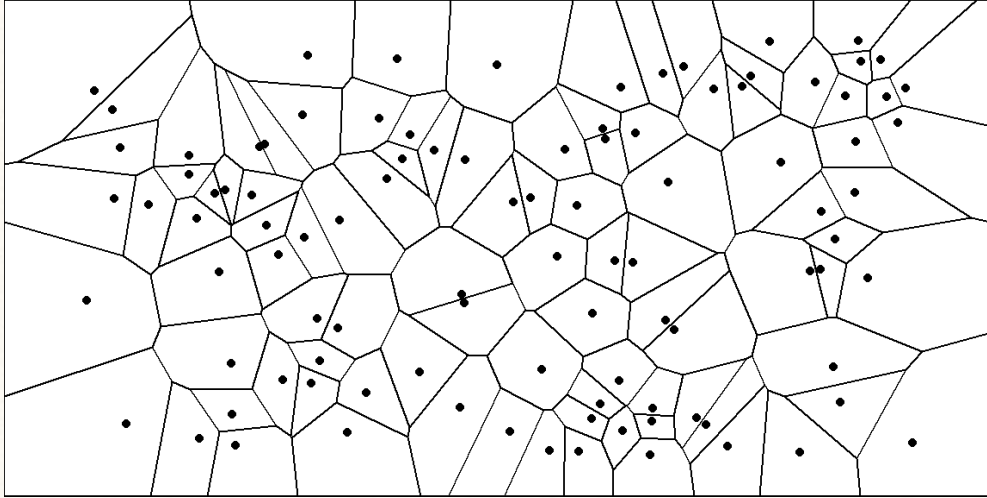
geometry,

...

Generalization

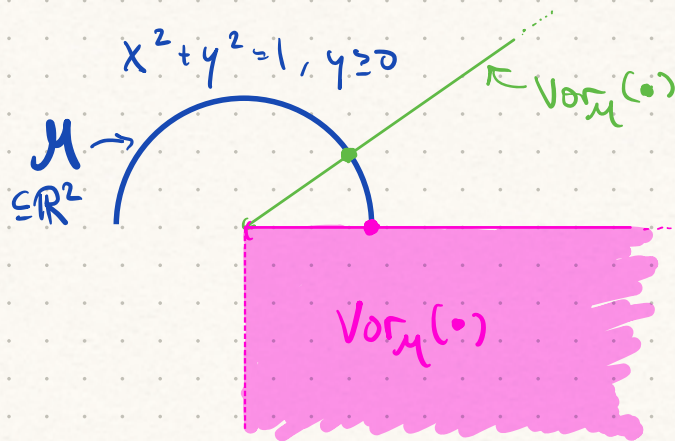
Why does the found solution function work well on new data?

Voronoi cells



For $S \subseteq \mathbb{R}^n$, the **Voronoi cell** at $p \in S$ is

$$\text{Vor}_S(p) := \{u \in \mathbb{R}^n \mid \forall q \in S, q \neq p: \|p - u\|_2 < \|q - u\|_2\}$$



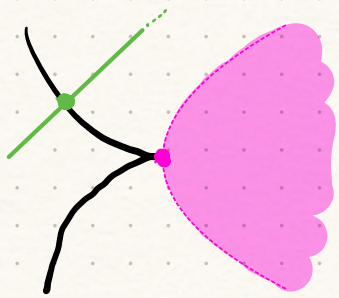
The **2 relative boundary points** are the only points on M with full-dimensional Voronoi cells!
 $\leadsto$ **implicit bias** towards ∂M

points in ∂M are global minima with positive probability on data u

singularities

$$y^2 + x^3 = 0$$

$$t \mapsto (-t^2, t^3)$$



$\Rightarrow$ implicit bias towards $\text{Sing}(\mathcal{M})$

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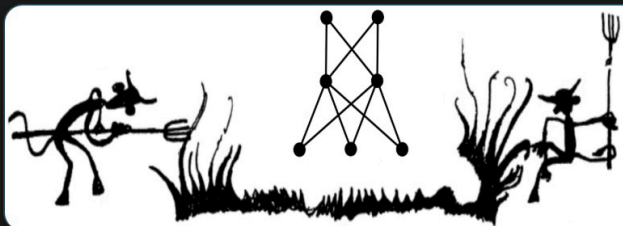
effect of singularity on overparametrization, ...



Neuroalgebraic Geometry

Auto

This page lists papers on the theory of algebraic neural networks.*



Algebra Unveils Deep Learning - An Invitation to Neuroalgebraic Geometry

Marchetti, Shahverdi, Mereta, Trager, Kohn

ICML Position Paper · 2025

[Read paper](#)

FILTERS

SORT BY

Activation

Architecture

Research Question

Year

Authors

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CNN Linear Learning Dynamics

The Riemannian Geometry associated to Gradient Flows of Linear Convolutional...

Achour, Kohn, Rauhut

AISTATS · 2026

[Read paper](#)

Polynomial Optimization Degree

Algebraic Robustness Verification of Neural Networks

Alexandr, Duan, Montúfar

preprint · 2026

[Read paper](#)

MLP Monomial Expressivity

Optimization Degree

Geometry and Optimization of Shallow Polynomial Networks

Arjevani, Bruna, Kileel, Polak, Trager

SIAM Journal on Applied Algebra and Geometry · 2026

[Read paper](#)